Module 14) Python – Collections, functions and Modules

7. Working with Dictionaries

• Iterating over a dictionary using loops

Iterating over a dictionary using loops is a fundamental concept in Python and other programming languages. Below is a theoretical explanation of how it's done, focusing on Python:

Why Iterate Over a Dictionary?

To:

- Access all keys or values
- Modify items
- Search or filter data
- Perform operations based on conditions

Ways to Iterate Over a Dictionary in Python

1. **Iterating Over Keys:**

```
for key in my_dict:
    print(key)  # prints each key
```

2. **Iterating Over Values**

```
for value in my_dict.values():
    print(value)
```

- ## Merging two lists into a dictionary using loops or zip().

Merging two lists into a dictionary is a common task in Python. This involves using one list for **keys** and another for **values**. There are two main ways to do this:

1. **Using zip()**

```
dict(zip(keys_list, values_list))
```

How It Works:

- The zip() function pairs elements from two lists into tuples.
- The dict() function then converts these tuples into key-value pairs.

2. **Using a Loop**

```
merged_dict = {}
for i in range(len(keys)):
    merged_dict[keys[i]] = values[i]
```

- ## Counting occurrences of characters in a string using dictionaries

Using a **dictionary** is one of the most efficient ways to count how many times each character appears in a string.

How It Works

1. **Initialize an empty dictionary.**
    - This will store characters as keys and their counts as values.
2. **Iterate through each character in the string.**
3. **For each character:**
    - **Check if the character already exists in the dictionary.**
        - If it does, increment its count.
        - If it doesn’t, add it to the dictionary with a count of 1.

Key Concepts

- **Dictionaries** allow fast lookups and updates, which makes them efficient for frequency counting.
- The logic is based on **conditional checking**:
    - Whether a character has been seen before (exists as a key).
- This method works for **any string**, regardless of its length or content.